Metformin in Rheumatoid Arthritis

Section 4.1 / 4.1.1 — working draft, 9 Oct 2026

4.1. Antidiabetic Drugs

Several drugs for type 2 diabetes also act on pathways that drive inflammation in RA, such as AMPK and NF-κB (Section 3). Insulin resistance is also common in RA and is linked to higher disease activity [Ormseth2013]. This section first covers metformin and pioglitazone, followed by glucagon-like peptide-1 receptor agonists (GLP-1RAs) and sodium–glucose cotransporter-2 (SGLT2) inhibitors.

4.1.1. Metformin

Metformin is an inexpensive first-line drug for type 2 diabetes and is generally safe [Gharib2021]. The drug appears to act mainly by activating AMP-activated protein kinase (AMPK), which in hepatocytes lowers acetyl-CoA carboxylase activity, increases fatty acid oxidation and suppresses lipogenic enzyme expression [Zhou2001]. Activated AMPK phosphorylates raptor, a step needed to inhibit mammalian target of rapamycin (mTOR) complex 1 (mTORC1) under energy stress [Gwinn2008]. Both mTOR complexes, mTORC1 and mTORC2, promote the differentiation of T helper 17 (Th17) cells [Nagai2013].

Animal studies consistently show that metformin reduces arthritis and protects the joint. In the collagen-induced arthritis (CIA) model (Table 1), immunized mice were given 5 mg/mouse of metformin orally three times per week for 9 weeks, starting 7 days after the first immunization. This treatment significantly reduced

arthritis severity and shifted the Th17 to regulatory T (Treg) cell ratio, with fewer Th17 cells and more Treg cells. Phosphorylation of signal transducers and activators of transcription 3 and 5 (STAT3 and STAT5) fell and rose, respectively, and osteoclast activity was reduced [Son2014]. Similarly, in rats with CIA, intraperitoneal (i.p.) metformin (100 mg/kg every 3 days for 3 weeks) protected the knee joints on micro-computed tomography (micro-CT) and on histopathology. Micro-CT showed preserved trabecular bone, histology showed less osteoclast formation and synovitis, and metformin also lowered markers of systemic inflammation, chondrocyte apoptosis and cartilage matrix degradation [Fan2020]. After body-surface-area conversion for a 60-kg adult [NairJacob2016], the oral dose of 5 mg/mouse three times weekly [Son2014] equals approximately 970–1,220 mg per dose, assuming a 20–25-g mouse because body weight was not reported. The intraperitoneal dose of 100 mg/kg every 3 days in rats [Fan2020] equals approximately 970 mg per dose, and both values lie within the usual clinical range of 500–2,550 mg/day [Naffaa2020].

Data from human joint tissue also suggest a link between AMPK and RA activity. In synovium from patients with RA, the active (phosphorylated) form of AMPK-α1 was higher than in synovium from osteoarthritis patients; in their serum, phosphorylated AMPK-α1 rose with the 28-joint Disease Activity Score (DAS28), and total AMPK-α1 rose with tumor necrosis factor-α (TNF-α) [Chen2020]. When fibroblast-like synoviocytes (FLS) from these patients were cultured, metformin at high

(millimolar) doses inhibited cell proliferation and migration and reduced TNF-α, interleukin (IL)-1β and IL-6, although the authors questioned whether such doses can be reached in patients [Chen2020].

Three clinical studies suggest a benefit of metformin as an add-on to conventional synthetic (cs) disease-modifying anti-rheumatic drugs (DMARDs). The larger of the two randomized trials, a double-blind study from Egypt, assigned 120 patients with active RA to metformin 1000 mg or placebo daily, each added to methotrexate 7.5 mg/week, over 12 weeks [Abdallah2021]. At 12 weeks, 80.8% of the metformin group reached an American College of Rheumatology 20% improvement (ACR20) response, compared with 54.7% of the placebo group ($P = 0.001$), without serious adverse effects in either group [Abdallah2021]. A randomized, single-blind trial enrolled 66 Egyptian patients with moderately or highly active RA, of whom 33 received metformin 850 mg twice daily in addition to csDMARDs and 33 received placebo. After 6 months, C-reactive protein (CRP) levels and DAS28-CRP scores were lower than with placebo, and only three metformin-treated patients withdrew because of gastrointestinal side effects [Gharib2021]. In a non-randomized prospective cohort study, 124 patients received low-dose metformin (250 mg twice daily) in addition to traditional csDMARD therapy for 90 days and were compared with 98 patients receiving csDMARDs only. Metformin use was associated with a faster fall in IL-6, TNF-α, high mobility group box 1 (HMGB1) and CRP than in the csDMARD-only group [Zhang2023]. A

meta-analysis of four randomized trials with 350 patients found that metformin lowered CRP after 3 months (mean difference −5.99 mg/L, 95% CI from −7.46 to −4.52) [Alshammari2026]. Over the same period, DAS28 also fell in the three trials that reported it (mean difference −0.80, 95% CI from −0.87 to −0.73) [Alshammari2026]. However, each of the four trials was conducted in Egypt or India, and two carried a high risk of bias because they were single-blind and did not report trial registration [Alshammari2026].

Data on RA prevention are conflicting. In a large UK cohort of people recently diagnosed with type 2 diabetes, RA risk was the same in metformin users as in users of other oral diabetes drugs (hazard ratio [HR] after adjustment 1.00, 95% CI 0.82 to 1.22) [Zemedikun2021]. A Mendelian randomization study suggested that metformin may protect against RA, but its genetic markers reflect metformin use and therefore largely track diabetes [Liang2023]. In an Israeli cohort of 113,749 people starting metformin, adherence was associated with a lower RA risk, which was lowest when 40–59% of days were covered rather than under 20% (adjusted HR 0.62, 95% CI from 0.45 to 0.84) [Naffaa2020]. In analyses stratified by sex, this association was limited to women [Naffaa2020]. Therefore, metformin may be a safe and inexpensive add-on to csDMARDs to lower RA disease activity. Whether it prevents RA is still unclear, and larger and longer double-blind randomized controlled trials (RCTs) that also check joint damage on X-ray are needed.